

Main

mapna mdigit

Asset tags: GTG • Tagsmg

Organisation

Company	mapna	Plant	mdigit
Location	plant	Operational status	active
Criticality	medium		

Identification

Manufacturer	smit	Model	220mw
Serial number	sm220gtsd12	Manufacturing year	2018
Installation date	5/8/2019	Commissioning date	8/9/2021

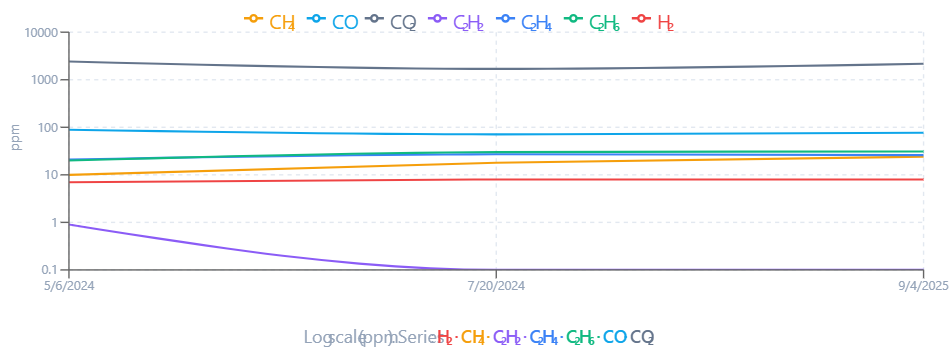
Electrical Nameplate

Type	GSU	Cooling	N/A
Vectorgroup	N/A	Windings	2
Rated power	220MVA	Forced cooling	200MVA
HW voltage	400kV	LW voltage	15.7kV
Tertiary voltage	N/A	Impedance	N/A
Frequency	N/A	Efficiency	N/A

Insulation & Construction

Oil type	N/A	Paper type	N/A
Insulation type	N/A	Insulation class	N/A
Oil volume	N/A	Weight	N/A
No load loss	N/A	Load loss	N/A
OLTC	No	Buchholz relay	Yes
Pressure relief	Yes		

Gas Concentration



Diagnostic Results by Test Date

Date	IEC	IEEE	Duv1	Duv4	Duv5	Duv6	Pen1	Pen2	IEC599	Rogers	Doern	ML
5/6/20...	Inves...	Inves...	T3	NA	NA	T3	T1	O	T2	T2	Ther...	D2
7/20/2...	Inves...	Inves...	T3	NA	NA	T3	T1	O	ND	UNK	Ther...	T2
9/4/20...	Inves...	Inves...	T3	NA	NA	T3	T1	O	ND	UNK	Ther...	T2

Recommendations

(To be completed by the reviewing engineer)

Validation: VALID

Total Tests

3

Valid Tests

2

Invalid Tests

1

Last Valid

2025-09-04

Final Decision

LOW

Conservative Clause 2

2-FAI (Furan)

N/A

NA

C900.3-C980

CQ (DGA)

2168 μl

LOW

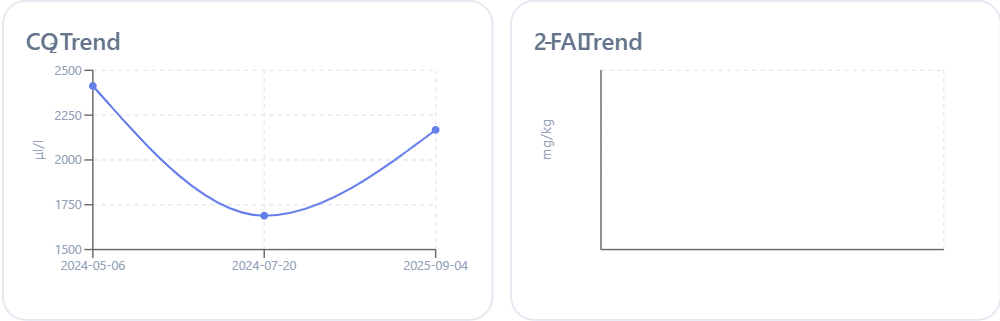
C90000-C98000

Rate of Increase

N/A

NA

Insufficient data for calculation



⚠ Warnings

- There are 2 valid CQ tests but no valid 2-FAI tests
- It is recommended to perform 2-FAI and CQ tests simultaneously with less than 6 months interval

Remaininlife

0.0rs

100%usedof30yrdesign

LifeConsumed

42.1yrs

100%ofdesign

CurrentHotSpot

92.0C

Topoil64.0Amb34.0

AgeinFAA

0.51

Load3.0pu

Method—DynamicGradient

Consumedlife0.0yrs

Remaininlife0.0yrs

AverageFAA0.839

Maxhotspot99.4C

Maxoil64.0C

Method—LoadCurrent

Consumedlife1121.42s

Remaininlife0.0yrs

AverageFAA517031.571

Maxhotspot293.1C

Lossat07.00

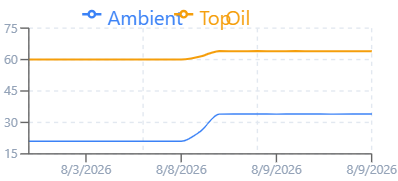
Agreement

6230076%.1

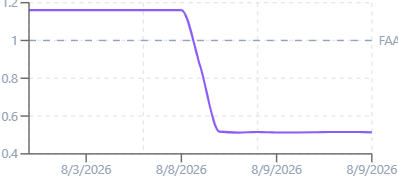
Criticalsignificantscrepancy

Δ1121.41sbetweenmethods

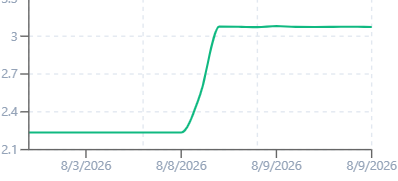
RawSignalsTopOil&Ambie(C)



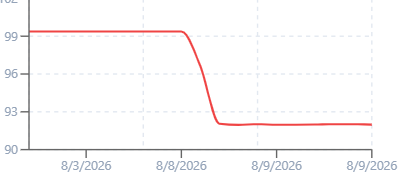
AgeinAccelerat(FAA)



Loadfact(pu)



HotSpotTemperat(C)



Recommendations

- ⚠️ Remaining life less than 5 years. Plan for replacement or major maintenance
- ✅ No acceleration being detected based on hot spot temperature
- ✅ Ageing rates within acceptable limits
- ⚠️ Overload detected at load factor 1.3 pu. Reduce load immediately